J^+

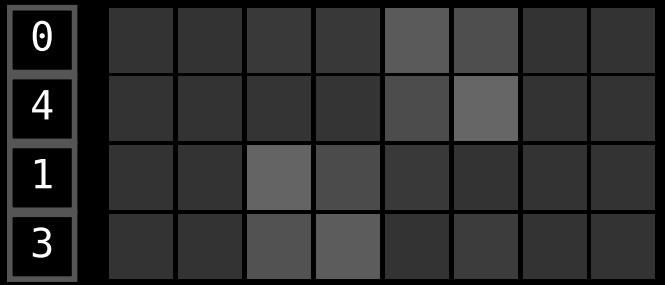
excitatory

$$\min_{W,\,H \geq 0} \|J^+ - WH^\top\|_F^2$$

nonnegative factors only add up: each is one part

$$J^+ \approx \sum_k \lambda_k w_k h_k^\top, \quad \|w_k\| = \|h_k\| = 1$$

λ_k: informativity, orders the factors



J^-

inhibitory

$$J^- \approx W^-(H^-)^\top$$

fit separately: the inhibitory factors